Client-Side ANN on the Web: A Wikipedia Search Demonstration

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Live search demo: https://wiki-ann-demo-20261022.pages.dev/client_wiki/
Clustering lab: https://wiki-ann-demo-20261022.pages.dev/client_wiki/benchmark.html
Code and benchmarks: https://github.com/DavidDenisov/ANN-on-the-Web

Abstract

We demonstrate browser-based approximate nearest neighbour (ANN) search over 35.2 million English Wikipedia passages hosted on a static CDN. The browser embeds questions locally, uses inverted file (IVF) routing to select vector clusters for parallel download, ranks their vectors, and fetches text for the top results. This avoids a full index download or an application search server. Visitors submit questions and inspect passages, timings, and download costs; a companion lab measures the recovery–bandwidth trade-off. On a desktop with primed CDN caches, at 32k clusters, SQ6 centroids, and four probes, a repeated fixed question returning ten passages averages 130.3 ms to worker reply, including embedding and text loading, with 3.25 MB of compressed payload. Worker startup averages 2.46 s. At the same settings, synthetic perturbations of L2 magnitude 0.4 yield 98.1% source recovery@1; this is not real-question recall. Queries and embeddings stay local, while file requests can reveal access patterns.

CCS Concepts

• **Information systems** → **Similarity search**.

Keywords

approximate nearest neighbors, semantic search, client-side retrieval

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1 Introduction

Semantic search on a static document collection usually requires a search service or a large local index. We demonstrate browser ANN over 35.2 million passages without either: local inverted file (IVF) routing selects remote cluster files for parallel download. The probe count controls the number of clusters fetched and ranked.

Our contribution is a static-Web deployment and an interactive view of established ANN techniques: search exposes per-question costs, while the lab isolates routing and centroid compression. The FP16 vectors occupy about 27 GB; each query transfers only selected clusters and result text from Cloudflare Pages.

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1.1 Related work

SPANN [1] keeps centroids in memory and posting lists on disk; we apply this partitioned-search pattern to static HTTP files. Tiptoe [4, Sections 3–5] also embeds queries and selects clusters locally, but its servers compute encrypted similarities over the corpus to hide cluster selection, followed by private URL retrieval. Our client downloads selected plaintext vector files and ranks locally, trading cryptographic privacy for static serving. Selective loading also appears in Pagefind's lexical search and sql.js-httpvfs's range-based database reads.

MeMemo [10] and WebANNS [6] use browser-local HNSW in RAM and IndexedDB; DiskANN [9] uses graph traversal. WebANNS evaluates up to 480,000 passages. Our larger remote corpus uses parallel cluster reads; the ANN algorithm is established.

2 Demonstration scenario

The two public pages use the same Wikipedia passage collection.

Wikipedia search. Visitors enter a question or other text and receive ranked passages with links to Wikipedia articles. Defaults are 32k clusters, SQ6 centroids, four probes, and ten results. Controls offer three cluster counts, six centroid formats, and 1–100 probes. The page reports embedding, centroid loading, retrieval, and text-loading times alongside vector downloads. The presenter first runs the example question in Figure 1, compares one and four probes, and inspects changed passages and byte counts. Visitors then try their own questions; these examples illustrate behaviour, not relevance scores. We keep the walkthrough to one or four probes to limit downloads; queries can be cancelled.

Clustering lab. The supplied lab (Figure 2) offers noise presets and separate checks for cluster count, probes, and centroid format. Visitors select 1–10,000 precomputed perturbed-passage queries per configuration (default 10) and can cancel runs. Short runs uniformly sample a shared subset; small samples illustrate behaviour. Paper measurements use all 10,000 queries, excluding embedding. Probe and compression checks use 16k clusters; the paper adds a 32k supplement. Full checks can take hours and transfer hundreds of GB. At the poster, visitors try short checks and interpret retained tables and the probe plot.

Public access. The onsite setup is a laptop running Chromium with Internet access and WebAssembly, plus a poster QR code for visitors' browsers. If connectivity fails, saved screenshots and retained plots support a narrated walkthrough without live search. The supplied source includes the client, builders, figures, compact measurements, and self-hosting instructions. Live hosting depends on the CDN; preview URLs are not a permanent archive.

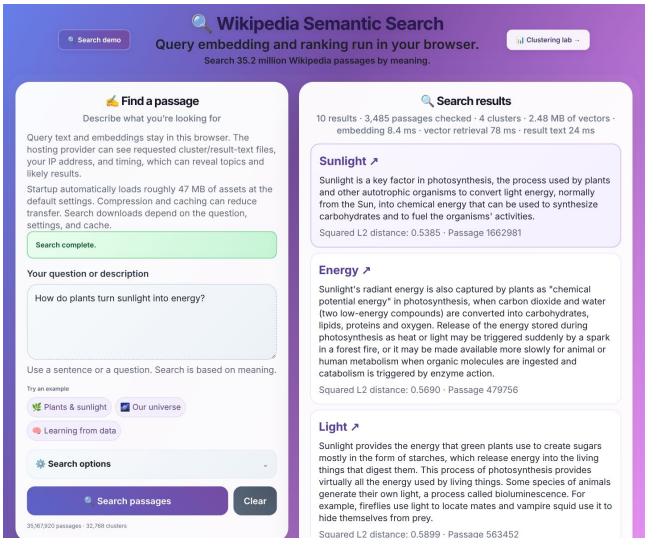


Figure 1: Search with privacy/download notices and passages. Capture timings are separate from the warmed averages in Section 4.3; typography is enlarged for print.

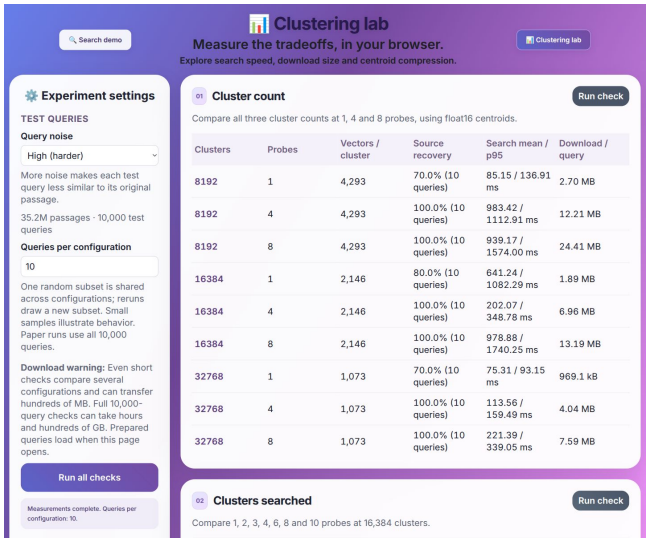


Figure 2: Lab after a 10-query-per-setting illustration, separate from paper measurements. Typography is enlarged for print.

3 Method outline

Figure 3 traces local computation and the static files downloaded at startup and during search.

The Maloyan dataset [7] provides the passages and their 384-dimensional `all-MiniLM-L6-v2` embeddings [8].

For queries, Transformers.js [3] loads the Xenova ONNX conversion [13] in `q8` format, truncating inputs to 256 tokens.

Index preparation. We train C centroids with Faiss k-means [2], using 20 iterations on about 2.1 million uniformly sampled passages. We assign each passage to its nearest centroid and store its ID and FP16 vector in that centroid's gzipped file. Centroids are published in FP16 and compressed formats; SQ6 uses six-bit scalar quantisation per component. Passage vectors remain FP16, occupying about 27 GB before compression.

Result text. Brotli files hold 500 contiguous passage IDs each, except the last. These text batches are shared across cluster counts. The browser fetches result batches and keeps decoded text in an 8 MiB cache.

Static hosting. The 70,336 text files use 71 branch deployments; vectors and centroids use 59 more in a second Pages project. Manifests map files to URLs. Our 1,000-file upload batches are a packaging choice, below the provider's 20,000-file Free-plan ceiling. Object storage can replace these hosts through manifest updates preserving CORS, content-encoding, and worker-isolation headers.

4 Supporting evaluation

All three indexes cover 35,167,920 passages. We test routing with synthetic perturbations and the ten-result worker workflow with a fixed question.

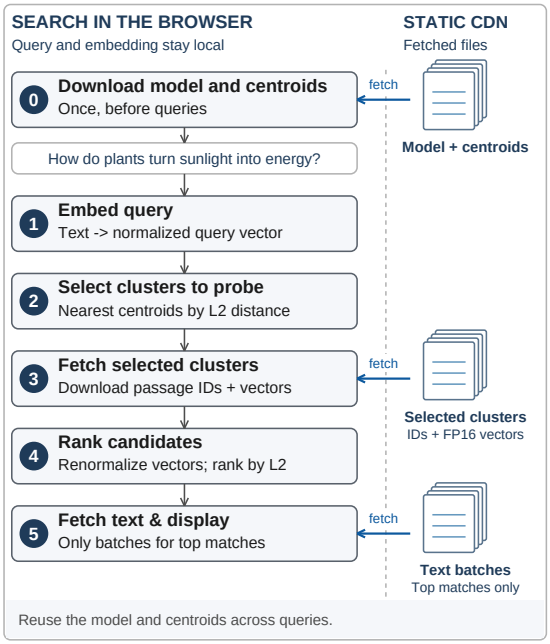


Figure 3: Local query embedding and ranking, with model and centroid downloads at startup and selective vector and text downloads per query.

Query setup. Configurations share 10,000 uniformly sampled distinct source passages. We add $v \in \mathbb{R}^{384}$ with uniformly random direction and $\|v\|_2 = \alpha = 0.4$ to each unit source vector, then renormalise. The lab also offers 0.05 and 0.15.

Perturbation scale and scope. On 1,000 sources, mean query-source L2 displacement is 0.378, or 55.5% of the mean nearest-neighbour distance (0.681) from an exact full-corpus scan excluding self matches. This controlled test exercises IVF boundary routing; isotropic passage noise does not model the question–passage distribution gap.

Metric. **Source recovery@1** is the fraction of perturbed queries whose source ranks first. Noise can favour another passage even in exact search; the plateau is not an ANN recall ceiling. Real-question recall@10, relevance, and q8 query-encoder effects remain unevaluated. For SQ6 at four probes, the 95% Wilson interval [12] is [97.77, 98.31]% (9,806/10,000), describing uncertainty under this synthetic protocol.

Test environment. Experiments use Ubuntu 22.04 on an Intel Core i5-12400F PC with 32 GB RAM and a GTX 1660 SUPER GPU. GPU Faiss prepared the indexes. Browser experiments use Chromium 155; inference uses Transformers.js 4.3.1 with CPU-only WebAssembly and four threads. Separate FAST.com checks reported about 300 Mbit/s download; CDN throughput was uncontrolled.

Embedding compatibility. Re-embedding 1,000 uniformly sampled passages with the q8 browser encoder yields mean and 5th-percentile cosine similarities of 0.9924 and 0.9892 to the stored vectors; this passage check does not validate question retrieval.

We compare $C \in \{8{,}192, 16{,}384, 32{,}768\}$ at 1, 4, and 8 probes, plus 2, 3, 6, and 10 at 16k and 32k. At 32k and four probes, we compare FP16, SQ8/SQ6/SQ4, symmetric INT8 [2], and PQ4 with 4-bit codes per eight-component block [5]. Cluster assignments and FP16 passage vectors stay fixed. Shared configurations run once.

Hosting and timing. Vector and text files are fetched before timing to prime the CDN cache; the fixed-question run uses 100 preliminary searches. CDN residency and connections remain uncontrolled; cache bypass does not establish a cold CDN. Timed searches fetch up to ten clusters in parallel. Timed fetches bypass browser HTTP caching, and text loads also bypass the decoded-text cache; untimed accuracy checks allow cached cluster files.

Retrieval time covers routing, cluster download, decompression, decoding, and ranking; Section 4.3 reports startup and complete-query costs. Centroids load once per configuration; separate download/decode benchmarks use 1,000 trials per file. Tables report mean and 95th-percentile times (p95), and gzipped vector payload in decimal MB (10^6 bytes). Configuration order was randomised; the original seed and shuffled supplement order are recorded.

4.1 Cluster count and probe budget

With FP16 centroids, four probes recover over 98% of source passages at all three cluster counts (Table 1). More clusters reduce vector downloads and retrieval time but increase the initial centroid load.

At 32k/FP16, increasing from four to eight probes raises source recovery from 98.1% to 98.8%, mean retrieval time from 135 to 240 ms, and vector payload from 3.98 to 7.78 MB. Figure 4 compares recovery and observed retrieval time; a companion payload plot is supplied with the source.

Table 1: FP16 four-probe retrieval at large noise: source recovery@1, mean / p95 time, and vector payload per query. 10,000 **queries per setting.**

Clusters	Probes	Recovery (%)	Time (ms)	MB/query
8192	4	98.4	462 / 662	13.65
16384	4	98.2	215 / 326	7.11
32768	4	98.1	135 / 182	3.98

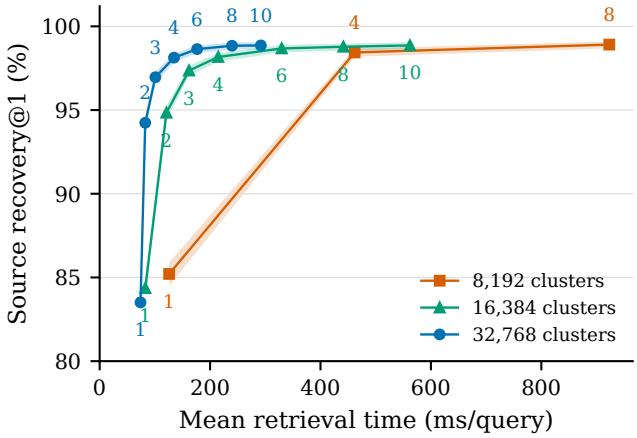


Figure 4: Recovery versus mean retrieval time (FP16, $\alpha = 0.4$; 10,000 queries/setting). Labels mark probes; shading joins pointwise 95% Wilson recovery intervals.

Table 2: Centroid formats at 32k clusters and four probes: source recovery@1 on 10,000 large-noise queries, file size, and loading mean / p95 over 1,000 trials per file. Decode includes gzip decompression.

Format	Recovery (%)	Size (MB)	Fetch (ms)	Decode (ms)
FP16	98.1	23.24	675 / 692	254 / 260
SQ8	98.2	11.23	300 / 339	113 / 115
SQ6	98.1	8.98	224 / 261	100 / 102
SQ4	97.4	4.86	83 / 102	88 / 91
INT8	98.1	10.99	290 / 342	103 / 106
PQ4	68.0	0.81	23 / 33	30 / 31

4.2 Centroid loading costs

At 32k clusters and four probes (Table 2), **SQ6 cuts mean centroid loading from** 929 **ms (FP16) to** 324 **ms**; both recover 98.1%.

FP16, SQ8, SQ6, and INT8 differ by at most 0.09 percentage points; paired significance was not assessed. SQ4 and PQ4 reduce recovery at the tested code sizes. SQ4 implies 468 Mbit/s versus a separate 300 Mbit/s check; throughput was uncontrolled. These synthetic results support 32k/SQ6 and four probes as a practical demo default within the tested grid; larger cluster counts remain unevaluated.

4.3 Client setup and query costs

Complete ten-result query. For "How do plants turn sunlight into energy?", the client returns ten passages at 32k/SQ6 and four probes. Across 1,000 repetitions, search submission to worker reply takes 130.3/150.3 ms (mean/p95), including embedding, vector retrieval, text loading, and message handling. Startup and DOM rendering are excluded; model and decoded centroids stay loaded. This repeated fixed-question p95 measures jitter, not variation across questions. The results require 2.48 MB of gzipped vectors and 0.77 MB of Brotli text (3.25 MB total payload, excluding HTTP headers).

Separate components. For synthetic queries, top-1 retrieval at 32k/SQ6 and four probes takes 122/187 ms (mean/p95), with 98.1% source recovery and 3.98 MB of vectors. The fixed question transfers about 38% fewer vector bytes; these component runs cannot be added to estimate varied-question end-to-end latency.

Startup. Startup spans worker creation until the model and 32k SQ6 centroids are ready, excluding page navigation. Across 100 fresh browser contexts (reusing the Chromium process), it takes 2.46/2.53 s (mean/p95). The model (23.7 MB), WebAssembly runtime (14.3 MB), and centroids (9.0 MB) total about 47 MB. Browser caching is bypassed; loading phases overlap.

4.4 Trade-offs and limitations

A hypothetical 48-byte/vector PQ index needs 1.69 GB for codes alone. Equating this with $47 + 3.25Q$ MB gives about 505 queries. This illustrates byte budgets, not equal-quality or latency baselines: IDs, codebooks, caching, and storage overhead change the crossing. One million uncached fixed-question searches transfer 3.25 TB before startup or headers. A server returns text alone but receives the query and needs search compute; that baseline is unmeasured.

FP16 passage vectors dominate transfer. SQ8/PQ passage codes with optional rescoring could reduce it, but change ranking and require evaluation. At 32k, manifest cluster sizes have median 1,023, p95 2,188, and maximum 8,263 passages. The four largest gzip files sum to 19.32 MB, an upper bound at four probes, not an observed query percentile; byte tails remain unmeasured. Varied real-question end-to-end mean/p95/max, cold-connection and cold-CDN trials, and throttled/mobile profiles remain necessary before generalising the desktop timings.

Privacy threat model. Assuming trusted client code, query text and embeddings stay local. The host/CDN can link IP addresses and timing to four of 32,768 cluster IDs and text-batch IDs. For uncached requests, intersecting these batches with downloaded candidate IDs can identify returned passages and potentially reconstruct the displayed result set; inference effectiveness is unmeasured. Displayed results do not establish which passages were read. While co-locating text with vector clusters would help mitigate this leakage, we did not implement this option for simplicity. Shared ID-based text batches let us vary the cluster count without maintaining a separate text layout for each count. Co-location would avoid result-specific text requests, although selected clusters would remain visible. PIR, as in Tiptoe [4], requires a different serving protocol.

5 Conclusion

Local IVF routing enables browser search over 35.2 million statically hosted passages. The demo exposes source recovery, latency, and bandwidth trade-offs.

6 Ethical use of data

We downloaded Maloyan's precomputed embeddings of 35,167,920 English Wikipedia passages (wikipedia-22-12-en) [7]. They were generated by applying the MiniLM encoder [8] to these passages. The encoder and its Xenova ONNX conversion [13] for browser queries are Apache-2.0 licensed. Result links credit Wikipedia contributors through source articles [11]. The supplied client and text exporters include a CC BY-SA 3.0 notice and `DATA-NOTICE.txt`, recording attribution and extraction/packaging changes. The embedding dataset specifies no separate licence. Passages reflect an archived snapshot and may be outdated or biased; linked articles may have changed or been deleted. Experiments involve no human participants.

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